Valuation Scheme

25900362

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) Regular EXAMINATION October 2025

THIRD SEMESTER

Discipline Specific Elective (DSE) - MG3DSEECT204 - SMART SENSORS WITH IOT

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Part A

Multiple Choice Questions
Answer all questions
Each question carries **1** marks

1. c) PIR sensor
2. b) Smart energy meters
3. b) Capacitive
4. a) Smartphone touch systems
5. a) Voltage from stress
6. b) Half the reference voltage
7. a) High temperature sensitivity
8. a) Linear Variable Displacement Transducer
9. c) The entire digital output is incorrect
10. c) To convert sensor signals into a compatible format
11. d) All of the above
12. b) Rotary actuator
13. b) Motor rotates at three-fourth of maximum speed
14. b) H-Bridge driver
15. d) All of the above
16. d) Application Programming Level
17. b) Electricity
18. b) Write API key
19. b) Smart sensor
20. b) ThingSpeak.writeField()

Part B

Short Answer Questions
Answer any **6** questions
Each question carries **5** marks

21. Why is filtering important in signal conditioning?

Write any 2 points 2.5marks each

1. **Noise Reduction:**
   Filters remove unwanted electrical noise and interference from sensor signals, improving measurement accuracy.
2. **Improved Signal Quality:**
   Filtering ensures that only the useful frequency components of a signal are retained, providing a cleaner and more reliable signal.
3. **Prevention of Aliasing:**.
   Before analog-to-digital conversion, filters (especially low-pass filters) prevent high-frequency components from causing aliasing errors.
4. **Protection of Electronic Components:**
   Filters protect sensitive devices like amplifiers and ADCs from sudden voltage spikes or high-frequency noise.
5. **Enhanced System Stability:**
   By eliminating unwanted fluctuations, filtering helps maintain consistent and stable operation in control and measurement systems.

22. List industrial applications of pressure sensors.

Write any 2 points 2.5marks each

1. **Hydraulic and Pneumatic Systems** – To monitor and control fluid or air pressure for safe and efficient operation.
2. **Process Industries (Oil, Gas, Chemical)** – To measure and regulate pressure in pipelines, reactors, and storage tanks.
3. **Automotive Industry** – Used in tire pressure monitoring systems (TPMS), fuel systems, and engine oil pressure monitoring.
4. **Water and Wastewater Management** – To monitor pump performance, detect leaks, and maintain proper pressure in pipelines.

23. What are the main criteria for selecting transducers?

Write any 5, 1mark each

1. **Range and Sensitivity:** The transducer should accurately measure within the required range and be sensitive enough to detect small changes.
2. **Accuracy and Linearity:** It must provide precise and linear output over its operating range to ensure reliable measurements.
3. **Response Time:** The transducer should have a fast enough response to capture changes in the measured quantity.
4. **Environmental Compatibility:** It should withstand environmental conditions

like temperature, humidity, vibration, and pressure in the application area.

5. **Stability and Repeatability:** The output should remain consistent over time and under repeated measurements.
6. **Physical Size and Mounting:** The transducer's size, shape, and mounting options should suit the system design.
7. **Output Signal Type:** The output (analog, digital, voltage, or current) must be compatible with the signal conditioning or control system.
8. **Cost and Reliability:** It should provide the best performance-to-cost ratio and have a long operational life with minimal maintenance.

24. Discuss one advantage and one disadvantage of active transducers?

One each, 2.5marks each

Advantages:

1. **No external power supply needed:**
   Active transducers generate their own electrical signal from the input energy (like heat, pressure, or motion), making them self-sufficient and easy to use in remote areas.
2. **Simple design and lower power consumption:**
   Since they don't require external excitation circuits, the overall system design is simpler and consumes less power.
3. **High reliability and portability:**
   Fewer components and no dependency on an external power source make these transducers more reliable and suitable for portable or field applications.

Disadvantages:

1. **Low output signal level:**
   The electrical signal generated is often weak and needs amplification before further processing.
2. **Limited to dynamic measurements:**
   Active transducers generally cannot measure static or steady-state quantities effectively (e.g., constant pressure or displacement).
3. **Environmental sensitivity:**
   Their performance can be affected by temperature changes, vibrations, or other environmental factors, reducing accuracy in harsh conditions.

25. How do linear actuators contribute to automation in the food and beverage industry? Give examples.

Contributions: (3.5marks)

1. **Precise Movement and Positioning:**
   Linear actuators enable accurate control of movement in filling, cutting, or sorting machines, ensuring consistent product quality and portion sizes.
2. **Automation of Repetitive Tasks:**
   They automate manual processes such as pushing, lifting, and adjusting components—improving efficiency, speed, and productivity.
3. **Improved Hygiene and Safety:**
   Electric and pneumatic linear actuators can be designed with **stainless steel and sealed housings**, suitable for **washdown environments**, ensuring compliance with

food safety standards.

4. **Energy Efficiency and Reduced Waste:**
   Their precise control minimizes material waste (e.g., during filling or packaging) and reduces downtime.

   Examples: (1.5marks)

   - **Bottling lines:** Controlling the position of nozzles for accurate liquid filling.
   - **Food packaging machines:** Sealing, cutting, and sorting packages with precision.
   - **Conveyor systems:** Adjusting guides or diverting products automatically.
   - **Mixing and dispensing systems:** Regulating ingredient flow or controlling valves.

26. Give three examples of relay applications in automobiles.

Explanation on relays, 2marks

Any 3 examples of relays, 1mark each

Headlight Control: Relays switch high-current headlights using a low-current dashboard switch.

Starter Motor Activation: Relays safely control the high current needed by the starter motor from the ignition switch.

Horn Operation: Relays allow the horn to receive sufficient current while protecting the horn button and wiring.

Cooling Fan Control: Relays turn on/off radiator or engine cooling fans based on temperature sensor signals.

Fuel Pump Control: Relays supply power to the fuel pump only when the ignition is on, ensuring safe operation.

27. Why is a relay module required for interfacing a water pump with ESP32?

Explanation on relays, 1marks

Explanation on ESP32, 1marks

Relay Module interfacing, 3marks

A **relay module** is required for interfacing a **water pump** with an **ESP32** due to the following reasons:

1. Current and Voltage Isolation:.
2. Ability to Switch High-Power Loads:
3. Safety and Protection:

28. Explain how industrial automation contributes to worker safety.

Any two, 2.5marks each

1. Reduces Human Exposure to Dangerous Tasks:

Automated machines handle **hazardous operations** such as handling chemicals, high temperatures, heavy loads, or toxic gases, keeping workers away from direct danger.

2. Minimizes Accidents Due to Human Error:

Automation ensures **consistent, precise, and controlled operations**, reducing

mistakes that can lead to injuries, equipment damage, or hazardous spills.

3. Remote Monitoring and Control:

Sensors and control systems allow operators to **monitor and control machinery remotely**, reducing the need for physical presence in risky areas.

4. Emergency Response and Safety Interlocks:

Automated systems include **safety interlocks, alarms, and emergency shutdown mechanisms** to quickly stop machines in case of faults or unsafe conditions.

5. Ergonomic Benefits:

Automation reduces **manual repetitive tasks**, heavy lifting, and awkward postures, lowering the risk of musculoskeletal injuries.

Part C

Essay Questions
Answer any **2** questions
Each question carries **10** marks

29. Discuss the need for sensors and transducers in modern industry with suitable examples.

Any two in detailed with examples, 5marks each

1. Process Monitoring and Control:

- Sensors continuously measure physical parameters like **temperature, pressure, level, flow, and humidity**.
- **Transducers** convert these physical quantities into electrical signals for processing.
- **Example:** In a chemical plant, a **temperature sensor** with a **thermocouple transducer** monitors reactor temperature to maintain safe and optimal reactions.

2. Automation and Efficiency:

- Sensors provide real-time data that allow machines to operate **automatically without human intervention**, reducing errors and increasing productivity.
- **Example:** In a bottling plant, **level sensors** detect liquid in bottles and trigger filling machines automatically.

3. Safety and Hazard Prevention:

- Sensors detect unsafe conditions and trigger alarms or shutdowns to prevent accidents.
- **Example: Gas sensors** in an oil refinery detect leaks and automatically activate ventilation or emergency systems.

4. Quality Control:

- Sensors ensure products meet required specifications, maintaining consistent quality.
- **Example: Optical sensors** in food processing detect defective items on a conveyor belt and remove them.

5. Energy Management:

- Sensors help monitor energy usage and optimize operations for efficiency.

- **Example: Current and voltage sensors** in factories track machine power consumption and prevent overloads.

30. Explain the advantages and limitations of LM358. Also list the applications of LM358.

Any 4 advantages and limitations, 4marks each
Any two applications, 2marks

31. Discuss the different types of linear actuators with examples.

Any 2 types, Working Principles, features and examples.. 5marks each
Electric Linear Actuators
Hydraulic Linear Actuators
Pneumatic Linear Actuators
Piezoelectric Linear Actuators
Magnetic or Linear Motor Actuators

32.Design a block diagram of an IoT-enabled automatic fan control system. Explain each component and its function in detail.

Explanation 2marks
Block Diagram 3marks
Components and Functions explanation 5marks